Impact of Noncontact Tactile Stimulation on Motion-Induced Electrodermal Responses and Subjective Motion Sickness

Elena N. Schneider, Caroline Lehser, Benedikt Buchheit, Mohamad Alayan, and Daniel J. Strauss

***Abstract*—Autonomous vehicles pose challenges regarding motion sickness (MS), as passengers engage in nondriving related tasks. To support the acceptance of self-driving cars, we evaluated effectiveness of a novel, noncontact tactile stimulation. The system provides anticipatory cues to passengers' palms using modulated ultrasound. The effectiveness of the system was determined using both, electrodermal responses (EDRs) evoked by driving maneuvers and subjective MS assessments. We observed significantly increased amplitudes of EDRs with higher MS levels, a relationship established for the first time in this context. The implementation of our stimulation system resulted in significantly reduced EDR amplitudes, indicating its efficacy in mitigating arousal and MS symptoms. This research contributes to developing objective MS evaluation methods and minimally intrusive mitigation strategies, potentially improving passenger comfort and acceptance of self-driving vehicles.**

***Index Terms*—Electrodermal response (EDR), motion sickness (MS), MS mitigation, self-driving cars, tactile stimulation.**

I. Introduction

THE future of the automotive industry will be defined by autonomous driving technology. However, this advancement presents new challenges, particularly the issue of motion sickness (MS) in self-driving vehicles. As passengers engage in non-driving related tasks (NDRTs), their eyes are no longer focused on the road, which can cause MS symptoms, such as headache, drowsiness, cold sweating, hypothermia, and nausea [1], [2]. In addition, innovative seating configurations that promote rearward movement can exacerbate the development of MS [3].

Received 11 March 2025; revised 13 August 2025; accepted 11 December 2025. Date of publication 14 January 2026; date of current version 3 April 2026. This work was supported by the German Federal Ministry of Research, Technology and Space, under Grant 03FH004IX5 (PI: DJS). This article was recommended by Associate Editor M. Dorneich. *(Corresponding author: Daniel J. Strauss.)*

Elena N. Schneider, Caroline Lehser, Benedikt Buchheit, and Daniel J. Strauss are with the Systems Neuroscience and Neurotechnology Unit, Faculty of Medicine, Saarland University & htw saar, 66421 Homburg, Germany, and also with the Center for Digital Neurotechnologies Saar, Faculty of Medicine, Saarland University, 66421 Homburg, Germany (e-mail: daniel.strauss@uni-saarland.de).

Mohamad Alayan is with ZF Friedrichshafen AG, 88046 Friedrichshafen, Germany.

This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/THMS.2025.3644434.

Digital Object Identifier 10.1109/THMS.2025.3644434

To address this challenge, various mitigation systems for MS have been proposed. In this study, we present a novel approach: A noncontact, mid-air tactile stimulation system that delivers information about upcoming driving maneuvers to participants' palms without direct skin contact, using ultrasound-modulated air. This system enables passengers to anticipate the vehicle's movements, potentially reducing MS symptoms.

To evaluate the effectiveness of this tactile countermeasure (CM), we utilized electrodermal responses (EDRs). Previous experiments by some of the authors have shown that EDRs occur after acceleration changes [4]. Therefore, the relation between MS, or CM, respectively, and EDRs is investigated here.

A. MS Theories and Susceptibility

Understanding the origin of MS is crucial to developing effective CMs. Two prominent theories attempt to explain the neural mechanisms underlying this phenomenon. The most widely accepted is the sensory conflict theory, which posits that MS is primarily the result of a mismatch between expected self-motion (based on visual input) and actual self-motion perceived through the vestibular system [5]. Another significant theory is the postural stability theory, which hypothesizes that low postural control, manifested as instability of the body or parts of the body, increases the risk of MS [6].

Although these theories provide insight into the circumstances that lead to MS, individual susceptibility remains a complex and not fully understood issue. It is established that anyone with a functioning vestibular system can experience MS [1], but some individuals appear to be completely resistant to its symptoms. The factors contributing to this variability in susceptibility have not yet been fully explained.

Given current limitations in understanding the causes of MS, researchers must explore alternative approaches to reduce the likelihood of developing MS symptoms or effectively counteract them when they occur. This ongoing investigation is crucial to improve comfort and acceptance in technologies prone to inducing MS, such as autonomous vehicles.

B. MS Mitigation

Pharmacological CMs for MS are well-known but often come with unwanted side effects such as drowsiness or blurred vision [7], [8]. Therefore, research into nonpharmacological CMs is steadily growing. Vitamin C and plant-based CMs such as

© 2026 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

ginger have been shown to generally have beneficial effects on MS (see [8]).

Behavioral CMs such as listening to favorite music [9] and sensory interventions, including stroboscopic lightning [10], [11], pleasant odors [12], and bone-conducted vibration [13], can help reduce MS symptoms or delay the onset of nausea. Other possibilities to minimize the risk of MS include visuospatial training [14], vibromotor reprocessing therapy [15], or in general, habituation to the stimulus [16]. In addition, permanent visualization of an artificial horizon [17], active head tilt [18], or systems that lead to similar head movements [19], can mitigate MS symptoms. Motion planning, including optimizing the driven route or adjusting driving style, can also have positive effects on MS [20], [21].

Besides the evaluation of these methods in virtual or simulated environments, some experiments are conducted in real-world driving, e.g., [14], [18], [19], [22], [23]. An aspect often investigated in real driving is the anticipation of upcoming dynamics, which has been shown to have positive effects on MS, e.g., [24], [25], [26], [27]. These anticipatory cues have been realized in different modalities that occur in most cases between 1 s and 3 s prior to different maneuvers. There have been visual cues, such as flashing arrows on a display prior to curves [23], [28], [29], auditory cues with different sounds depending on the upcoming acceleration [30], [31], haptic cues, like vibrating wristbands [32] or vibration realized on the seat, e.g., [26], [27], [33], and also the combination of visual and haptic cues [22], [34], [35]. Most of these works show a reduction in MS scores. Especially the usage of haptic CMs can be beneficial, because driving in a vehicle is often combined with a loud sound environment and a visual task, which could influence visual or auditory CM.

CM systems that inform users about the upcoming driving dynamics must consider both effectiveness and intrusiveness. That means that these systems should not distract passengers from their NDRT. One goal of this study was to evaluate the effectiveness of the haptic device as an MS CM (using virtual haptics). In this approach no physical contact is made with the participant, as vibrations are generated through modulated ultrasound. These nonintrusive devices are ultrasonic boards that generate haptic sensations in mid-air (see Section II-D). In [36], a device that generates haptic sensations in mid-air with ultrasound was introduced to provide haptic feedback on top of interactive surfaces. The system creates constructive interference patterns of focused ultrasound as focal points in mid–air. One great advantage of this method is that you do not have to put on extra systems, and it can be sensed through clothes, which can be advantageous regarding possible application sites. This method can be used for a contactless transfer of information from the car to the passenger. In [37], it was already used in the context of a human–machine interfaces in vehicles to provide contactless haptic feedback while gesture detection or while using a touchscreen, but not as a CM for MS.

C. Objective MS Metrics

Since the evaluation of MS is often based on subjective information or questionnaires, which can worsen MS symptoms [38], objective MS evaluation methods are of great relevance. Various MS models and the ISO2631-1 standard [39] can calculate MS levels based solely on accelerations, head position, and motion perception [40], [41], [42], [43]. We propose that a more individualized method for objectively assessing MS could be highly advantageous. Given the numerous factors influencing susceptibility to MS, such as sleep duration, food intake, and initial health conditions, and general resistance to MS, a method based on individual physiological data is highly relevant. Incorporating all of these aspects into a purely model-based approach can be challenging. Furthermore, for a future mitigation system that can be activated or deactivated based on the current objectively evaluated MS state of passengers, reliable MS detection is crucial.

Many measurements have already been performed to find correlations between physiological data and MS. In particular, the use of machine learning algorithms that include various physiological correlates shows very promising results in the classification of MS [44], [45]. Some experiments that record physiological data, performed during real car drives, show a correlation between MS and physiological parameters [4], [45], [46], [47], [48]. For the evaluation of anticipatory CM, only a few studies investigated changes in physiological data during real driving, mainly heart rate (HR) and heart rate variability (HRV) [32], [34], [49], [50], [51].

Several different markers have been used so far for MS detection such as electrogastrography (EGG) [47], [52], electroencephalography (EEG) [53], HR [46], [54], HRV [34], [55] or electrodermal activity (EDA) [4], [46], [47], [48], [56].

EDA is used very commonly to evaluate the activity of the autonomous nervous system and psychophysiological states [57]. Cold sweating, a common symptom of MS, contributes to an increase in EDA [5], [48]. There are two different parts of the EDA that are used to analyze sympathetic activity. The first is the tonic EDA, the slow-changing part, mostly used in MS experiments, and the phasic EDA, which consists of the fast-changing signal. The phasic EDA consists of single peaks, EDR, which can be nonspecific EDRs (NS.EDRs), which occur spontaneously, and evoked EDRs, which are elicited by arousing stimuli. These are used in various areas of psychology [58]. The frequency of NS.EDRs has been shown to be correlated with sympathetic activity, whereas evoked EDRs have been observed to exhibit alterations in amplitude.

In the 1970 s, it was already investigated that the frequency of NS.EDRs correlates with the number of braking events [59] and the number of curves [60], but this has not been further investigated. The peak amplitude of the evoked EDRs is known to correlate with the valence and arousal of the stimulus [57], while the latency is not well addressed in the literature [61]. Some research demonstrated that arousal and anticipation can influence the latency of EDRs [61], [62], with [61] finding that greater awareness leads to shorter latency.

Since HR and HRV did not show effects related to MS or CM [32], [34], [49], [50], while tonic EDA demonstrated a relationship with a visual CM [51], EDA was chosen as the biomarker in this study. In addition, efforts to estimate EDA without contact [63], which would offer clear benefits

in-vehicle use, further supported this choice. Previous experiments by Smyth et al.[48] and Mühlbacher et al. [47] presented correlations between tonic EDA and MS at the group level, but do not recommend using tonic EDA as a reliable MS classification due to diverse individual results. Significant increases in tonic, as well as in phasic EDA with higher MS levels have already been found on an individual basis by [46]. They also found that phasic EDA, unlike tonic EDA, is not influenced by driving time or time on task. This makes it useful for comparing EDA trends according to MS rather than time on task. However, elicited responses (EDRs) as part of the phasic EDA, have not been evaluated regarding the aspect of MS so far. Earlier research by the authors showed that driving maneuvers (speed bumps, accelerations in *x*-, and *y*-direction) increase the EDA on a short time scale, comparable to EDRs [4]. To build on those findings, we intended to evoke clear EDRs to different driving maneuvers and investigate the relation between EDR parameters and subjective motion sickness level (SMSL) and between EDRs and the presented contactless haptic CM. Evoked EDRs can play a beneficial role in future MS detection in self-driving vehicles, as reactions on a short time scale can accelerate the evaluation.

II. Materials and Methods

A. Participants and Experimental Protocol

A total of 29 volunteers were recruited for this experiment (18 male, age: 25.4 ± 2.6 years). Six participants had to be excluded due to synchronization errors and data loss for the analysis of psychophysiological data, resulting in a group of 23 participants (14 male, age: 25.5 ± 2.8 years). For the evaluation of subjective and behavioral data, no data had to be excluded, which led to a sample size of 29 participants. Each participant performed two trials on two different days (one condition per trial/day), spaced at least 14 days apart, to ensure complete recovery from any MS symptoms and to avoid habituation to the driven route [64], as the route was driven similarly for both trials. The order of the two conditions was randomized: a haptic CM system and a placebo CM control trial, explained in Section II-D. In the first trial, participants signed an informed consent form and completed forms complying with laboratory hygiene guidelines related to COVID-19. They completed a questionnaire on MS susceptibility consisting of eight questions from [65], [66], [67]. This questionnaire yielded a susceptibility score between zero (resistant) and 12 (extremely susceptible), with an average score of 7.3 for all participants. The questionnaire can be found in the supplementary material and is explained in more detail in [68]. Before each trial, participants confirmed a normal state of health.

In both trials, the participant was seated in the middle rear seat of a passenger car (VW Touran 2nd gen. 2017, Type: 1 T/CJSAAF, 1.8TSI, 132 kW), completing a driving paradigm designed to induce MS symptoms (see Fig. 1). After the participants were properly seated in the car, a baseline of 3 min was recorded in the parked vehicle, followed by a second baseline recorded for 5 min while the car was driven with a velocity of 10 km/h without any strong maneuvers. Although the baseline data was not used for normalization purposes for the presented data, it was done to acclimate the participant to the car and its temperature and to compare physiological data between driving with a normal head position and a lowered gaze, which was the head position from the start of the actual measurement.



Fig. 1. Participant seated on the middle rear seat of the test-vehicle during haptic condition (a, top) with palms placed above two UH boards to both sides of the passengers, and during control condition (b, bottom) with two lights illuminating the footwell to both sides as placebo CM. The monitor for the NDRT is placed in the middle in front of the participant. EDA electrodes are positioned on the nondominant hand of the participant on both conditions.

The NDRT for the participant involved reading a text and watching silent landscape videos on a small monitor in a loop, changing every 5 min. A foot switch was used to scroll the text, as the hands were used for haptic stimulation (see Section II-D). For comparability between trials, the foot switch was also used in the control condition to reduce hand movement, which can distort the EDA signal. Every 5 min, participants were asked to verbally report their current MS symptoms (fatigue, nausea, sweating, headache, or dizziness) and their SMSL on a scale of zero to 10, comparable to the MS misery scale [69]. Instead of a description of the symptoms, a visual analog scale with cliparts was used for a better intuitive and faster evaluation for the participants while driving. The scale can be found in the supplementary material. Cancellations due to severe symptoms were always marked with a 10 in the SMSL. The duration of the measurement was about 30 min, including the baseline drive. The 5-min interval for the SMSL rating was selected to achieve a balanced ratio between time resolution and deflection of the NDRT. After the measurement, the participants answered a few questions about the text they read to control their attention to it and about their subjective perception of the CM, the tasks, and the driven route. All participants were aware that they could stop the measurement at any time without giving a reason. All procedures were carried

out in accordance with the Declaration of Helsinki and were approved by the local ethics committee (Identification number: 181/18; 149/23 Ärztekammer des Saarlands; Medical Council of the Saarland).

B. Data Acquisition

We recorded physiological data with commercially available wireless sensors from Shimmer Research Ltd. (Ireland). Electrocardiography (ECG)(Shimmer3 ECG Unit), Pulse on the index finger, EDA, with dry Ag/AgCl electrodes, on the middle and ring finger (Shimmer3GSR + Unit), and skin temperature (T) on the little finger of the nondominant hand (Shimmer3 BridgeAmplifier). In the second trial, we additionally attached pregelled disposable EDA electrodes on the forehead 2 cm below the hairline of the participants. In this article, only the EDA data collected on the finger of the participants are analyzed. Physiological data were collected with a sampling frequency of 25 6Hz. The participants were continuously monitored with a camera setup, which contained two stereo near-infrared (NIR) cameras (850 nm, XIMEA xiQ, XIMEA GmbH, Germany) and one thermal imaging camera (FLIR AX5, FLIR Systems, Inc., USA). The analysis of the camera data is beyond the scope of this article. Driving dynamics data were recorded from the vehicle's controller area network (CAN), including GPS-based position, steering wheel angle, indicator flags, internally calculated velocities based on wheel speed and angle, angular velocity, and acceleration from the standard sensors in-built in a series-produced VWTouran 2nd gen. Furthermore, a VBOX 3i (Racelogic LTd, U.K.), which contains an external GPS/GLONASS dual antenna system and an inertial acceleration and angle measurement unit was used with a sampling frequency of 100 Hz.

C. Route Design and Maneuver Classification

The driven route consisted of 12 laps, each approximately 500 m in length, resulting in an overall driving distance of 6.1 km. The route was taken in a closed area. Each lap consisted of one of three round types, each including different maneuvers. All laps were driven at an average speed of 20 km/h. Round one (R1) involved an acceleration up to 40 km/h and a slowdown to 10 km/h. The second round (R2) included a short slalom section and two u-turns. The last round (R3) consisted of two short stop-and-go maneuvers, with acceleration up to 30 km/h and slowing to 10 km/h. Furthermore, each round included two speed bumps (driven at 10 km/h), braking to a standstill, and a left-right-left (LRL) combination of curves (exact description of the rounds in Fig. 2). The maximum accelerations in the x and y directions were about 5.5 $\mathrm{m/s^2}$. The different rounds were driven in the order R1–R2–R2–R3 three times in a row for every participant. Participants were asked about possible road habituation after the experiment. Only a few participants were able to predict some of the u-turns, as they had an overall orientation on the road, knowing when the end was reached and the car had to be turned. However, exact prediction without additional information from the CM was not possible for the participants.

All maneuvers were driven by a pretrained driver, separating the vehicle control actions to isolate the steering and acceleration effects (no braking while steering/no steering while braking), to prevent different accelerations acting in parallel, which could lead to overlapping EDRs and influence the data analysis, as it is not exactly known if EDA reacts similarly to each acceleration. Markers were placed on the ground to ensure reproducibility of the route. An exact profile of the driving dynamics on the complete route can be found in [70]. The motion sickness dose values (MSDV) per trial for each participant can be found in the Supplementary Material. These results show no differences between the driving in both trials. For the analysis of single maneuvers and to detect the exact time points entering a maneuver, we used a combination of the GPS position of the car on the predefined route, the acceleration and the wheel speed of all four wheels. The driven route resulted in 24 speed bumps, 12 braking points, 14 curves, 12 LRL, 12 u-turns, six slaloms, three strong accelerations, and six stop and go maneuvers per participant. If participants aborted the measurement, there were fewer maneuvers carried out.

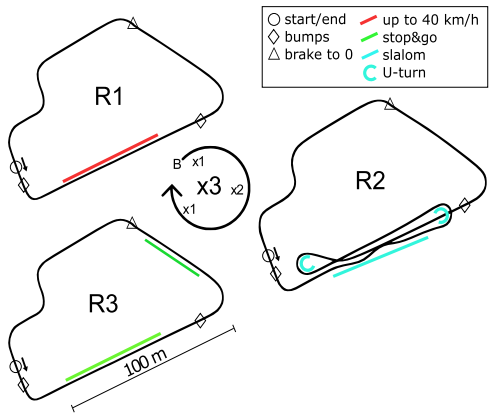


Fig. 2. Three different round types (R1, R2, R3) with included maneuvers. R1 included a strong acceleration up to 40 km/h (red line), R2 included slalom maneuvers and u-turns (cyan lines) and R3 consisted of two stop and go maneuvers (green lines).

D. Countermeasures

As a CM, two ultrasound arrays (Ultrahaptics Stratos Explore, Ultraleap, England) (UH boards) were used. These arrays use modulated ultrasound to achieve tactile stimulation in mid-air (see [36]). The ultrasound boards consist of an array of 16×16 ultrasonic transducers, which generate focal points in the air. The ultrasound at 40 kHz was modulated by amplitude with a sine wave of the modulation frequency to change the intensity at the focal point. A frequency of 200 Hz was used as the modulation frequency, which is in the range to which mechanoreceptors in the skin are sensitive [71]. The generation of focal points also causes an acoustic noise that was indistinguishable from the motor sound of the car. The focal points were generated 20 cm above the transducer array, so the distance between the palm and the UH boards had to be kept constant during driving to generate the best stimulus intensity. To maintain this stable distance, the hands were placed on hand rests on both sides of the participants [see Fig. 1(a)], with markers for the fingers, allowing participants to find the correct position even after moving their hands. Participants were also asked not to move their hands during the drive.

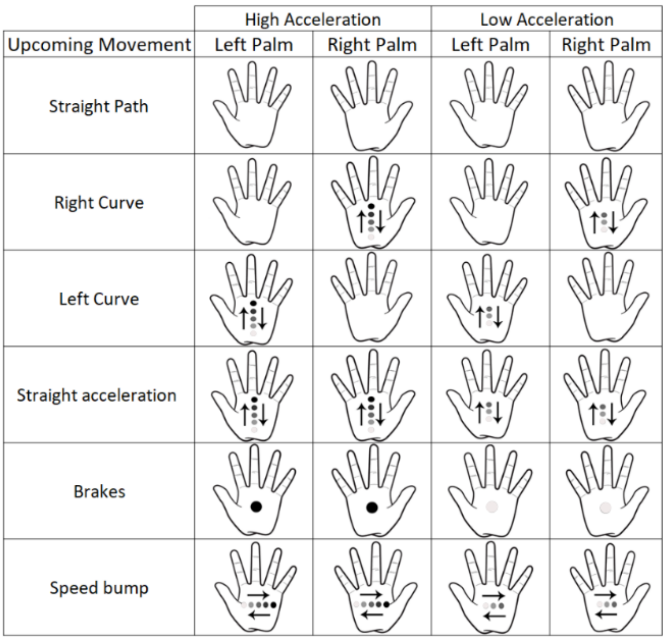


Fig. 3. Focal point presentation of the haptic CM related to different driving maneuvers. Stimulation intensity is represented by the color intensity, i.e., light grey stands for light intensity and black points indicate maximal intensity.

Depending on the upcoming maneuver, different stimuli were presented to the participants with the two boards. They did not receive any description of the stimuli used or their relation to the driving route, as the study aimed to investigate subliminal and subconscious information transport to the participants. This setup allowed the main focus to be on the NDRT and tested whether a possible future CM system could work from scratch or whether training on a specific CM would always be necessary to gain benefits. For right/left curves, the right/left hand was stimulated. Acceleration, braking, and driving through speed bumps were presented in both hands in parallel with different intensity and spatial changes of the focal point (see Fig. 3). There were three different intensity levels, with accelerations of 2 m^2 and speed bumps always depicted with the highest intensity. GPS supported the car position mapping process on the predefined route to detect upcoming driving maneuvers and stimulate 1 s prior to the actual maneuver. Before each haptic condition, it was checked whether the boards were functioning correctly and it was tested that each participant could distinguish between the different stimuli, which was the case for everyone. This did not take longer than 5 min. In addition, immediately after measurement, it was tested that the UH boards were still intact.

The placebo system used for the control condition consisted of two LED bars (Cameo Pixbar 500 Pro) slowly changing light intensity and color in a randomized order, positioned to illuminate the footwell of the car on both sides of the passenger [see Fig. 1(b)]. There was no relation to any driving dynamics of the car, so no effect on MS is assumed. However, a placebo system was used to create similar psychological situations for both conditions, as it is already known that MS can be influenced by changes in psychological situations [38]. The hands of the participants were held relaxed and in contrast to the haptic condition, were not positioned on the hand rests, to have an unobstructed view on the lights.

E. Data Analysis

Physiological data were analyzed using MATLAB 2022b (The MathWorks, Inc). The raw EDA data were low-pass filtered with a second order butterworth filter with a cutoff frequency of 10 Hz and visually inspected to remove artifacts. The data were normalized to $\mu S/\mathrm{cm}^2$ regarding the diameter of the electrodes. EDA was divided into the tonic and phasic part with a convex optimization algorithm by [72]. For the analysis presented here, only the phasic EDA was used.

For the analysis of the EDRs, the EDA signal from $t = 0$ s to $t = 20$ s after each driving maneuver was normalized by subtracting the mean of the EDA signal over $t = -5$ s to $t = 0$ s prior to this event, to be able to examine each event separately, without the influence of spontaneous responses shortly before an event. Maneuvers such as LRL or stop and go have a duration of about 13–17 s and EDA responds with latency up to 5 s, therefore, the interval after an event was set to 20 s.

The amplitudes (EDRamp) and latency (EDRlat) of the EDRs were automatically registered by detecting the responses in time windows between t = -2 s and t = 8 s relative to the onset of the event for short maneuvers, such as brake points, curves, u-turns, and speed bumps, which have a duration of about 3 s. For other maneuvers, responses with durations from 8 s to 17 s were detected from the onset of the event $t = 0$ s up to $t = 10$ s. Under laboratory conditions, EDRs to stimuli, such as emotional triggers, typically have latencies ranging from 1 s to 3 s. We adapted these intervals because there is no exact starting point for driving maneuvers. Slight changes in velocity or acceleration can already be perceived by the human body. For example, easing off the accelerator before entering a curve can initiate the curve and make it perceptible even before the steering wheel is turned. Due to these uncertainties, we selected relatively large time intervals for the detection of EDRs. For longer maneuvers, which can last up to 15 s, multiple maneuvers are often combined into one. For instance, LRL consists of three curves, stop, and go includes acceleration and braking, and the slalom involves more than one turning point. This can lead to overlapping EDRs, which cannot be accurately discriminated here. Therefore, we considered the interval up to 10 s s after event onset to most likely detect the EDR corresponding to the first part of the maneuver combination. Automatic detection was supported by visual inspection. This analysis results in one EDRamp and one EDRlat value per maneuver per participant, resulting in 99 values per participant for EDRamp and EDRlat, respectively. This number of values can be reduced when the participant aborted the measurement or not every maneuver elicits an EDR visible in the depicted time windows.

F. Statistical Analysis

For statistical analysis we used R (version 4.2.2, 2022 the R Foundation for Statistical Computing) and $lme4$ [73] to perform a linear mixed effects analysis of the relationship between

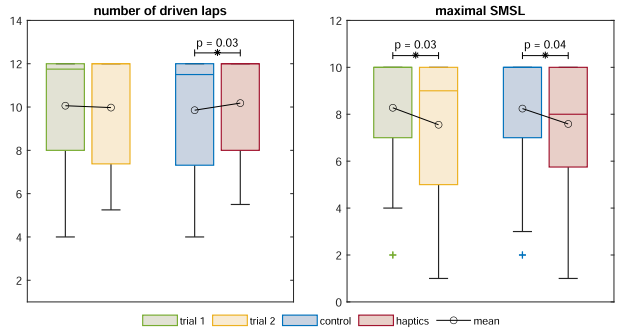


Fig. 4. Boxplots of SMSL and number of driven rounds divided by trial number (left) and separated by condition, haptic CM (red) and control condition (blue) (right).

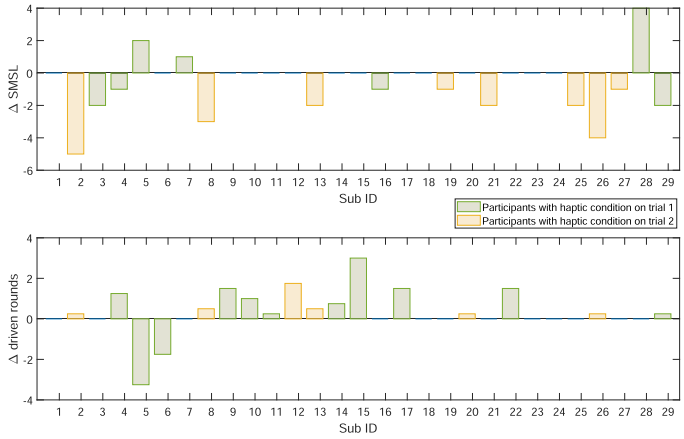


Fig. 5. Top: Difference between SMSL of the control condition and the haptic condition of all individuals. Negative values depict a lower SMSL in the haptic condition and positive values the other way around. For 14 participants there was no difference in the SMSL for the two conditions. Bottom: Difference between number of driven laps for the control condition and the haptic condition of all individuals. Positive values indicate higher number of driven laps in the haptic condition. 12 participants drove the same number of laps in both conditions. Green bars indicate that the haptic condition was performed on the first trials for these subjects, and the yellow bars indicate, that the haptic condition was performed on the second trial.

EDRamp and CM, EDRlat and CM, EDRamp and SMSL, and EDRlat and SMSL. For the analysis of the relationship between EDRamp/EDRlat and CM, we used the condition (haptic or placebo), the SMSL (0–10) and trial number (first or second trial) as fixed effects. As random intercepts, we added the participant ID (1–23) and the driving maneuver (braking point, curves, LRL, u-turn, slalom, acceleration, speed bump, stop, and go). We took the logarithmic transform of EDRamp as predictor variables to account for linearity. The residuals did not show obvious deviations from homoscedasticity or normality. Colinearity between SMSL and CM was ruled out. The p-values were obtained by likelihood ratio tests of the full model described above against a model without the questioned effect, CM or SMSL.

Subjective and behavioral data were checked for significant differences using the Wilcoxon signed rank test for paired data with no normal distribution. Subjective data were the maximum SMSL rated by participants. As a behavioral measure, we used the completely driven number of laps before the measurement was aborted. The maximum number of laps driven is 12.

III. Results

A. Objective Data

Results of the linear mixed-effect model showed that EDRamp is significantly increased with higher SMSL ($\chi^2(1) = 15.41, p < 0.0001$), while EDRlat did not show significant dependency on SMSL ($\chi^2(1) = 3.12, p = 0.08$). The haptic CM lowered the EDRamp significantly by about 0.16 μs ($\chi^2(1) = 28.83, p < 0.0001$) while there was no significant effect on EDRlat ($\chi^2(1) = 3.03, p = 0.08$). The trial number (first or second trial) did not have an influence on both the EDR parameter, EDRamp ($\chi^2(1) = 1.49, p = 0.22$) and EDRlat ($\chi^2(1) = 0.84, p = 0.36$).

B. Subjective and Behavioral Data

There were 12 out of 29 participants who aborted the measurement during the haptic condition (41% dropout rate), compared to 18 abortions during the control condition (62% dropout rate). The number of fully driven laps was significantly increased for the haptic condition compared to the control condition and SMSL was significantly decreased (see Fig. 4). Fig. 5 shows the difference between the maximal SMSL of the haptic condition and the control condition of each participant. A total of 14 of the 29 participants had the same maximum SMSL under both conditions. A total of 11 of these participants had to abort the measurement under both conditions. Only three of 29 individuals showed a higher SMSL in the haptic condition compared to the control condition, resulting in 12 participants with a higher SMSL in the control condition. Taking a look at the development of SMSL according to the trial number (first drive or second drive after 14 days), the SMSL significantly decreased in the second trial. However, the number of completely driven rounds was not altered between the two trials (see Fig. 4).

After each trial, participants were asked if they realized that the CM was working during the drive and if they could get any information from it. Only nine of 29 participants were able to identify the different stimuli of the CM and connect some of them to driving maneuvers. Eight of these nine participants showed lower SMSL values in the haptic condition compared to the control condition. Five subjects did not recognize stimulation during the drive at all (although it was successfully tested before and after the measurement).

IV. Discussion

A. Noncontact Haptic CM

The objective of this study was to evaluate an ultrasound-based, noncontact, haptic system for mitigating MS using EDRs as a new reliable MS measure, after showing that EDRs are related to MS. The haptic system operated at an almost subliminal level and was evaluated using subjective, behavioral, and objective data. The results indicate a mitigating effect with a lower SMSL and an increase in the driven distance. A haptic CM was selected due to the independence of tactile stimulation

from the sound environment and lighting conditions, which can vary rapidly within a vehicle. In addition, the haptic modality is seldom occupied by NDRTs, as passengers typically engage their visual or auditory modalities while reading or listening to music.

While previous experiments have indicated positive effects of anticipatory haptic CMs [22], [32], the current study explores additional aspects. First, the CM utilized here is based on modulated ultrasound, enabling contactless operation and the ability to be perceived through light clothing [74]. This feature is advantageous for future applications where the CM's position may vary, eliminating the need for additional devices such as vibrating wristbands used in [22].

Somatosensory evoked potentials are similarly triggered by the presented system and actual haptic sensations, demonstrating comparable perception of stimuli on the hand [75]. However, perceiving stimulation in areas other than the palm remains challenging [76]. Although the face is sensitive to stimulation, ultrasound near the ears can be harmful [76]. Given that hands are frequently involved in NDRTs, further research on the stimulation itself is necessary before integration into vehicles.

B. Effectiveness of Subliminal Character

The second aspect, rarely investigated in the literature (e.g., [64]), is the subliminal character of an anticipatory CM. In contrast to other research on anticipatory cues that mitigate MS, the participants in this experiment did not receive any information about the functionality of the CM. Participants were informed that MS could occur and that systems to mitigate its development were being tested. All information from the UH boards on upcoming driving maneuvers and the timing of these maneuvers was something the participants had to discern on their own. Testing the subliminal character of a CM is relevant for future applications, as it allows the system to be activated without additional explanation, facilitating its use. However, only a few subjects (9 out of 29) were able to identify the different stimuli of the CM and connect some of them to driving maneuvers. Of these nine participants, eight showed a lower SMSL in the haptic condition compared to the control condition. Among the other 20 participants, only four showed lower SMSL. This indicates that understanding the working mode of the CM, or habituation to the system, is necessary to improve its effectiveness. The visual system tested in [64] did not show strong effects, further supporting the need for better explanation of the working mode.

Another aspect, often overlooked in other experiments, except for [64] and closely related to the almost subliminal working mode, is the placebo effect. As MS is highly sensitive to psychological factors [77], having two conditions — one with CM and one without — can strongly influence the development of MS. Simply knowing that a CM system is being tested can decrease subjective MS evaluation due to the placebo effect (e.g., [78], [79]). To eliminate the influence of a placebo effect, we aimed to establish the same psychological condition in both trials by using a placebo CM with randomly changing lights as the control condition. We assume this could be the reason for the relatively weak statistical differences compared to other experiments such as [31], which do not mention using a placebo condition. We encourage the use of a placebo condition when testing MS mitigation systems to ensure the effect is based on the system itself and not solely on psychological factors. However, prior to conducting our experiment, we did not assess whether the implemented lights condition (used here as the control condition) might elicit a placebo effect when compared to the absence of any additional stimulus.

In addition to the placebo effect, other aspects can influence MS, which vary between trials. Although we attempted to conduct measurements at the same time of day to account for similar hormone levels, such as cortisol [80], which can influence MS [81], factors such as food intake, hours of sleep, and emotional state of subjects, known to influence MS, could not be adequately controlled.

C. Order Effect

Regarding differences between trials, it is important to note that the relevance of the results is diminished by an order effect in the SMSL, despite randomizing the order of conditions. SMSL is significantly reduced using the haptic CM, but there is also a significant reduction in SMSL in the second trial. This could indicate that the SMSL for the haptic condition is only reduced because the SMSL is lower in the second condition. To rule out this assumption, we also checked other parameters, such as the number of driven laps. Using the CM, participants drove significantly more laps. This was not the case in the second trial compared to the first trial, indicating that the CM had a higher impact than the trial number. In addition, the objective markers, EDRamp, showed a significant decrease in the haptic condition, but no significant change related to the trial.

Several factors could contribute to the reduction in SMSL in the second trial. In addition to general nervousness in the first trial, for first time participants, the self-evaluation of MS susceptibility was always done prior to the first trial, which could result in the so-called priming effect [82]. Simply taking this questionnaire can increase the risk of higher SMSL scores, as reported by [38]. Another factor that can promote an order effect is habituation to the paradigm. In [64] and [22], order effects were reported with one to 17 and three days, respectively, between measurements. To reduce the risk of habituation, a minimum of 14 days between the two conditions was implemented here. However, an order effect in the subjective data is still visible.

For future experiments, we suggest ensuring the absence of priming effects and habituation by, for example, using different subject groups for testing different conditions or increasing the time between two trials to reduce the probability of order effects in subjective data. These noted effects again highlight the importance of objective evaluation procedures, which are less likely to be influenced by factors such as the placebo effect [78].

D. Objective Evaluation

In addition to the subjective evaluation of the CM, one goal of this study was to use EDR parameter as an objective marker for MS and CM. As a secondary finding a higher EDR amplitude (EDRamp) was associated with higher SMSL. Furthermore, a

significant increase in EDRamp was observed in the control group, indicating a positive effect on MS and on sympathetic arousal using the haptic CM.

In the context of driving, EDRs have previously been investigated only in drivers, for example, as a response to sudden events such as the appearance of a pedestrian [83], or as a response to movements involved in braking or steering [59]. These events can be excluded as triggers for EDRs in the present experiment, as participants had no view of the road ahead and did not interact physically or emotionally with the car or traffic. In the context of fully autonomous vehicles, where there is no driver, EDRs could support advanced driver assistance systems in accurately detecting MS, activating CMs according to the passenger's MS level, and automatically adjusting the intensity if necessary. Since we demonstrated that the haptic CM significantly reduced SMSL and that EDRamp is influenced by MS, the reduced EDRamp is likely a result of the lower SMSL due to the CM. However, it remains unclear to what extent anticipation or MS affects EDR. Measurements in fully resistant participants would be required to determine whether increasing amplitudes occur only in susceptible individuals, thus serving as an objective measurement method for MS. The general effectiveness of the system compared between susceptible and resistant participants would also be of high interest. In this study, all participants showed susceptibility to MS. However, significant changes in EDR were only visible when using the individual as a random factor. These results align with similar findings regarding tonic EDA [46], highlighting the individual variability of EDA [58].

Incorporating the individual variability of EDA, as well as differences in MS susceptibility, symptom development, and physiological responses to driving maneuvers, into promising MS classification algorithms based on machine learning tools [44], [45], could enhance MS detection and lead to more reliable results. Implementing such a system in self-driving vehicles with personalized profiles, combined with a haptic CM, could adjust the CM to the passenger's current state, thereby improving comfort during autonomous travel.

E. Limitations

Several limitations associated with the proposed paradigm should be considered when interpreting the results. The primary factor contributing to MS during highly automated driving is the NDRT, which divert visual attention from the road and result in sensory conflict. The choice of task can also influence MS symptoms, so both reading and video-watching tasks were included. The text was kept informative with no emotional content to minimize immersion, making the task more comparable between participants. The task changed every 5 min to a landscape video with many perspective changes and fast motion, leading to increased MS symptoms for some participants. We cannot completely exclude the possibility that the videos led to additional visually induced MS (VIMS) and that the videos from the first and second trials contributed to VIMS to different degrees. If the video presented in the first trial included slightly more motion, it could have caused the order effect mentioned above. Due to these additional stimuli that induce VIMS in some subjects, the effectiveness of the CM may be diminished. This is further supported by the fact that, while using the UH boards, fewer participants reported that the reading task was more uncomfortable. This indicates that the boards had a higher effect during reading due to the additional visual stimuli during the video, which were not covered by the CM stimulation. To eliminate such issues, we suggest using only one NDRT during the measurement drive to guarantee a constant load for the participants.

An additional limitation is the differing arm position of the participants in both conditions. According to postural instability theory, alterations in posture (including arm position) can also influence MS. Although the hands were not secured to the boards, there was a lack of arm support in one condition compared to the other, which may reduce the beneficial impact of haptic CM. This aspect must be considered in subsequent research.

One last limitation is the automated detection of EDRs. Although there are some automated detection algorithms (e.g., [72], [84], [85]), it is still difficult to find relevant responses in a paradigm generating many stimuli and therefore overlapping responses, as it is the case in our paradigm (different driving maneuvers, uneven road conditions, NDRT performance, and MS). Especially for maneuvers consisting of more than one acceleration, like slalom or stop-and-go, there is more than one response, which has to be separated. The EDR detection was supported by visual inspection to provide the best possible detection, but it cannot be completely guaranteed that each EDR selected is solely triggered by the prior maneuver.

In addition to the limitations stated above, this study demonstrated a significant correlation between EDRamp and SMSL. Furthermore, a tactile, noncontact anticipatory CM was implemented, which was capable of significantly reducing EDRamp, increasing the driven distance, and decreasing the SMSL.

V. Conclusion

The major goal of this study was to introduced an ultrasound-based, haptic, contactless stimulation system designed to provide anticipatory cues to passengers in a vehicle, driven using the "Wizard of Oz" method. Our results show that this system effectively reduces MS symptoms and increases the distance driven before the measurement has to be aborted. As a secondary finding, we demonstrated the potential of EDRs as an objective measure to assess MS, showing increased EDR amplitudes with higher SMSL and reduced EDR amplitudes when using the CM. The integration of MS detection with the presented CM holds promise for developing an advanced assistance system in self-driving cars, offering high user acceptance due to its simplicity. Future research is essential to validate these findings and ensure the system's effectiveness in real-world traffic conditions.

References

[1] K. E. Money, "Motion sickness," *Physiol. Rev.*, vol. 50, no. 1, pp. 1–39, 1970.

[2] E. Nalivaiko, J. A. Rudd, and R. H. So, "Motion sickness, nausea and thermoregulation: The "toxic" hypothesis," *Temperature*, vol. 1, no. 3, pp. 164–171, 2014.

[3] S. Salter, C. Diels, P. Herriotts, S. Kanarachos, and D. Thake, "Motion sickness in automated vehicles with forward and rearward facing seating orientations," *Appl. Ergonom.*, vol. 78, pp. 54–61, 2019.

[4] E. N. Schneider et al., "Electrodermal responses to driving maneuvers in a motion sickness inducing real-world driving scenario," *IEEE Trans. Human-Mach. Syst.*, vol. 52, no. 5, pp. 994–1003, Oct. 2022.

[5] M. Treisman, "Motion Sickness: An evolutionary hypothesis," *Science*, vol. 197, no. 4302, pp. 493–495, 1977.

[6] G. E. Riccio and T. A. Stoffregen, "An ecological theory of motion sickness and postural instability," *Ecological Psychol.*, vol. 3, no. 3, pp. 195–240, 1991.

[7] F. Schmäl, "Neuronal mechanisms and the treatment of motion sickness," *Pharmacol.*, vol. 91, pp. 229–241, 2013.

[8] G. Rahimzadeh et al., "Nutritional and behavioral countermeasures as medication approaches to relieve motion sickness: A comprehensive review," *Nutrients*, vol. 15, no. 6, Art. no. 1320, 2023.

[9] B. Keshavarz and H. Hecht, "Pleasant music as a countermeasure against visually induced motion sickness," *Appl. Ergonom.*, vol. 45, no. 3, pp. 521–527, 2014.

[10] C. M. Webb, A. Estrada, and J. R. Athy, "Motion sickness prevention by an 8-Hz stroboscopic environment during air transport," *Aviation Space Environ. Med.*, vol. 84, no. 3, pp. 177–183, 2013.

[11] M. F. Reschke, J. T. Somers, and G. Ford, "Stroboscopic vision as a treatment for motion sickness: Strobe lighting vs shutter glasses," *Aviation Space Environ. Med.*, vol. 77, no. 1, pp. 2–7, 2006.

[12] B. Keshavarz, D. Stelzmann, A. Paillard, and H. Hecht, "Visually induced motion sickness can be alleviated by pleasant odors," *Exp. Brain Res.*, vol. 233, pp. 1353–1364, 2015.

[13] S. Salter, C. Diels, S. Kanarachos, D. Thake, P. Herriotts, and D. A. Depireux, "Increased bone conducted vibration reduces motion sickness in automated vehicles," *Int. J. Hum. Factors Ergonom.*, vol. 6, no. 4, pp. 299–318, 2019.

[14] J. Smyth, P. Jennings, P. Bennett, and S. Birrell, "A novel method for reducing motion sickness susceptibility through training visuospatial ability–a two-part study," *Appl. Ergonom.*, vol. 90, 2021, Art. no. 103264.

[15] E. Molefi, R. Palaniappan, and I. McLoughlin, "Motion sickness reduction through vibro-motor reprocessing therapy: A first study," in *Proc. 10th Int. IEEE/EMBS Conf. Neural Eng.*, 2021, pp. 937–940.

[16] C. D. Wood et al., "Habituation and motion sickness," *J. Clin. Pharmacol.*, vol. 34, no. 6, pp. 628–634, 1994.

[17] D. Tal et al., "Artificial horizon effects on motion sickness and performance," *Otol. Neurotol.*, vol. 33, no. 5, pp. 878–885, 2012.

[18] T. Wada and K. Yoshida, "Effect of passengers' active head tilt and opening/closure of eyes on motion sickness in lateral acceleration environment of cars," *Ergonomics*, vol. 59, no. 8, pp. 1050–1059, 2016.

[19] V. Erusu, S. Kuhn, M. Sama, S. Hwang, and J. Raiti, "Queasy: A tilting interface to alleviate vehicle-induced motion sickness in passengers," in *Proc. IEEE Glob. Humanitarian Technol. Conf.*, 2022, pp. 150–153.

[20] Z. Htike, G. Papaioannou, E. Siampis, E. Velenis, and S. Longo, "Fundamentals of motion planning for mitigating motion sickness in automated vehicles," *IEEE Trans. Veh. Technol.*, vol. 71, no. 3, pp. 2375–2384, Oct. 2022.

[21] M. R. Siddiqi, S. Milani, R. N. Jazar, and H. Marzbani, "Motion sickness mitigating algorithms and control strategy for autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 304–315, Jan. 2023.

[22] J. Karjanto, N. M. Yusof, M. Z. Hassan, J. Terken, F. Delbressine, and M. Rauterberg, "An on-road study in mitigating motion sickness when reading in automated driving," *J. Hunan Univ. Natural Sci.*, vol. 48, no. 3, 2021.

[23] R. Hainich, U. Drewitz, K. Ihme, J. Lauermann, M. Niedling, and M. Oehl, "Evaluation of a human–machine interface for motion sickness mitigation utilizing anticipatory ambient light cues in a realistic automated driving setting," *Information*, vol. 12, no. 4, 2021, Art. no. 176.

[24] C. Diels and J. E. Bos, "Self-driving carsickness," *Appl. Ergonom.*, vol. 53, no. 2015, pp. 374–382, 2015.

[25] O. X. Kuiper, J. E. Bos, E. A. Schmidt, C. Diels, and S. Wolter, "Knowing what's coming: Unpredictable motion causes more motion sickness," *Hum. Factors*, vol. 62, no. 8, pp. 1339–1348, 2020.

[26] A. J. Reuten, J. E. Bos, M. H. Martens, J. Rausch, and J. B. Smeets, "Mitigating motion sickness by anticipatory cues," *Vibration*, vol. 7, no. 4, pp. 1266–1278, 2024.

[27] N. Jalgaonkar, D. Schulman, A. Kim, J. Chen, and S. Awtar, "Preemptive haptic stimuli to mitigate car sickness and its influence on task performance: A pilot study," in *Proc. IEEE 26th Int. Conf. Intell. Transp. Syst.*, 2023, pp. 5014–5019.

[28] P. Feenstra, J. E. Bos, and R. N. van Gent, "A visual display enhancing comfort by counteracting airsickness," *Displays*, vol. 32, no. 4, pp. 194–200, 2011.

[29] J. Bin Karjanto et al., "Situation awareness and motion sickness in automated vehicle driving experience: A preliminary study of peripheral visual information," in *Proc. 9th Int. Conf. Automot. User Interfaces Interactive Veh. Appl. Adjunct*, 2017, pp. 57–61.

[30] O. X. Kuiper, J. E. Bos, C. Diels, and E. A. Schmidt, "Knowing what's coming: Anticipatory audio cues can mitigate motion sickness," *Appl. Ergonom.*, vol. 85, 2020, Art. no. 103068.

[31] J. Maculewicz, P. Larsson, and J. Fagerlönn, "Intuitive and subtle motion-anticipatory auditory cues reduce motion sickness in self-driving cars," *Int. J. Hum. Factors Ergonom.*, vol. 8, no. 4, pp. 370–392, 2021.

[32] N. B. M. Yusof, "Comfort in autonomous car: Mitigating motion sickness by enhancing situation awareness through haptic displays," Ph.D. dissertation, (Research TU/e / Graduation TU/e), Industrial Design, Eindhoven Univ. Technol., Eindhoven, Netherlands, 2019.

[33] A. Reuten, J. Smeets, J. Rausch, M. Martens, E. Schmidt, and J. Bos, "The (in) effectiveness of anticipatory vibrotactile cues in mitigating motion sickness," *Exp. Brain Res.*, vol. 241, no. 5, pp. 1251–1261, 2023.

[34] J. Karjanto, N. M. Yusof, J. Terken, F. Delbressine, and M. Rauterberg, "Level of motion sickness based on heart rate variability when reading inside a fully automated vehicle," *Mech. Eng. Soc. Ind.*, vol. 2, no. 2, pp. 72–81, 2022.

[35] N. B. M. Yusof, J. Karjanto, M. Z. Hassan, J. Terken, F. Delbressine, and M. Rauterberg, "Reading during fully automated driving: A study of the effect of peripheral visual and haptic information on situation awareness and mental workload," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19136–19144, Oct. 2022.

[36] T. Carter, S. A. Seah, B. Long, B. Drinkwater, and S. Subramanian, "Ultrahaptics: Multi-point mid-air haptic feedback for touch surfaces," in *Proc. 26th Annu. ACM Symp. User Interface Softw. Technol.*, 2013, pp. 505–514.

[37] K. Harrington, D. R. Large, G. Burnett, and O. Georgiou, "Exploring the use of mid-air ultrasonic feedback to enhance automotive user interfaces," in *Proc. 10th Int. Conf. Automot. User Interfaces Interactive Veh. Appl.*, 2018, pp. 11–20.

[38] S. D. Young, B. D. Adelstein, and S. R. Ellis, "Demand characteristics in assessing motion sickness in a virtual environment: Or does taking a motion sickness questionnaire make you sick?," *IEEE Trans. Visual. Comput. Graph.*, vol. 13, no. 3, pp. 422–428, May/Jun. 2007.

[39] International Organization for Standardization (ISO), "Mechanical vibration and shock - evaluation of human exposure to whole body vibration - part1: General requirements, standard ISO 2631-1," Geneva, Switzerland, 1997.

[40] J. Bos and W. Bles, "Modelling motion sickness and subjective vertical mismatch detailed for vertical motions," *Brain Res. Bull.*, vol. 47, no. 5, pp. 537–542, 1998.

[41] N. Kamiji, Y. Kurata, T. Wada, and S. Doi, "Modeling and validation of carsickness mechanism," in *Proc. SICE Annu. Conf.*, 2007, pp. 1138–1143.

[42] T. Wada, "Computational model of motion sickness describing the effects of learning exogenous motion dynamics," *Front. Syst. Neurosci.*, vol. 15, 2021, Art. no. 634604.

[43] B. Buchheit, E. N. Schneider, M. Alayan, F. Dauth, and D. J. Strauss, "Motion sickness prediction in self-driving cars using the 6DOF-SVC model," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 13582–13591, Aug. 2021.

[44] B. Keshavarz, K. Peck, S. Rezaei, and B. Taati, "Detecting and predicting visually induced motion sickness with physiological measures in combination with machine learning techniques," *Int. J. Psychophysiol.*, vol. 176, pp. 14–26, 2022.

[45] R. Tan et al., "Motion sickness detection for intelligent vehicles: A wearable-device-based approach," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst.*, 2022, pp. 4355–4362.

[46] T. Irmak, D. M. Pool, and R. Happee, "Objective and subjective responses to motion sickness: The group and the individual," *Exp. Brain Res.*, 2020, Art. no. 0123456789, doi: 10.1007/s00221-020-05986-6.

[47] D. Mühlbacher, M. Tomzig, K. Reinmüller, and L. Rittger, "Methodological considerations concerning motion sickness investigations during automated driving," *Information*, vol. 11, no. 5, Art. no. 265, 2020.

[48] J. Smyth, S. Birrell, R. Woodman, and P. Jennings, "Exploring the utility of EDA and skin temperature as individual physiological correlates of motion sickness," *Appl. Ergonom.*, vol. 92, no. 2021, Art. no. 103315, 2020. doi: 10.1016/j.apergo.2020.103315.

[49] J. Karjanto, N. Md Yusof, C. Wang, J. Terken, F. Delbressine, and M. Rauterberg, "The effect of peripheral visual feedforward system in enhancing situation awareness and mitigating motion sickness in fully automated driving," *Transp. Res. Part F: Traffic Psychol. Behav.*, vol. 58, pp. 678–692, 2018. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1369847818300913

[50] N. M. Yusof, J. Karjanto, J. Terken, F. Delbressine, and G. Rauterberg, "Gaining situation awareness through a vibrotactile display to mitigate motion sickness in fully-automated driving cars," *Int. J. Automot. Mech. Eng.*, vol. 17, no. 1, pp. 7771–7783, 2020.

[51] E. Kim, H. Gil, J. Ryu, I. Oakley, and G. Shin, "The effect of peripheral cues on motion sickness mitigation when using a VR HMD in a car," in *Proc. Hum. Factors Ergonom. Soc. Annu. Meeting*, Los Angeles, CA, 2022, pp. 1967–1971.

[52] M. J. Cevette et al., "Electrogastrographic and autonomic responses during oculovestibular recoupling in flight simulation," *Aviation, Space, Environ. Med.*, vol. 85, no. 1, pp. 15–24, 2014.

[53] Y. Wei, Y. O. Okazaki, R. H. So, W. C. Chu, and K. Kitajo, "Motion sickness-susceptible participants exposed to coherent rotating dot patterns show excessive N2 amplitudes and impaired theta-band phase synchronization," *NeuroImage*, vol. 202, 2019, Art. no. 116028, doi: 10.1016/j.neuroimage.2019.116028.

[54] S. R. Holmes and M. J. Griffin, "Correlation between heart rate and the severity of motion sickness caused by optokinetic stimulation," *J. Psychophysiol.*, vol. 15, no. 1, 2001, Art. no. 35.

[55] L. T. LaCount et al., "Dynamic cardiovagal response to motion sickness: A point-process heart rate variability study," in *Proc. 36th Annu. Comput. Cardiol. Conf.*, 2009, pp. 49–52.

[56] H. Wan, S. Hu, and J. Wang, "Correlation of phasic and tonic skin-conductance responses with severity of motion sickness induced by viewing an optokinetic rotating drum," *Perceptual Motor Skills*, vol. 97, no. 3 II, pp. 1051–1057, Dec. 2003.

[57] W. Boucsein, *Electrodermal Activity*. Berlin, Germany: Springer, 2012.

[58] W. Boucsein et al., "Publication recommendations for electrodermal measurements," *Psychophysiology*, vol. 49, no. 8, pp. 1017–1034, 2012.

[59] M. Helander and B. Hagvall, "An instrumented vehicle for studies of driver behaviour," *Accident Anal. Prevention*, vol. 8, no. 4, pp. 271–277, 1976.

[60] D. Taylor, "Drivers'galvanic skin response and the risk of accident," *Ergonomics*, vol. 7, no. 4, pp. 439–451, 1964.

[61] R. Sjouwerman and T. Lonsdorf, "Latency of skin conductance responses across stimulus modalities," *Psychophysiol.*, vol. 56, no. 4, 2019, Art. no. e13307.

[62] C. Van Oyen Witvliet and S. R. Vrana, "Psychophysiological responses as indices of affective dimensions," *Psychophysiol.*, vol. 32, no. 5, pp. 436–443, 1995.

[63] M. Bhamborae, E. N. Schneider, P. Flotho, A. Francis, and D. J. Strauss, "Lumeda: Image luminance based contactless correlates of electrodermal responses," *Physiol. Meas.*, 2024.

[64] D. Bohrmann, A. Bruder, and K. Bengler, "Effects of dynamic visual stimuli on the development of carsickness in real driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 5, pp. 4833–4842, May 2022.

[65] J. T. Reason and J. J. Brand, *Motion Sickness*. Cambridge, MA, USA: Academic Press, 1975.

[66] J. F. Golding, "Motion sickness susceptibility questionnaire revised and its relationship to other forms of sickness," *Brain Res. Bull.*, vol. 47, no. 5, pp. 507–516, 1998.

[67] J. F. Golding, "Predicting individual differences in motion sickness susceptibility by questionnaire," *Pers. Individual Differences*, vol. 41, no. 2, pp. 237–248, 2006.

[68] B. Buchheit, T. Grün, E. N. Schneider, M. Alayan, and D. J. Strauss, "Motion sickness prediction based on passenger's self evaluation," in *Proc. IEEE 26th Int. Conf. Intell. Transp. Syst.*, 2023, pp. 925–930.

[69] J. E. Bos, S. N. Mackinnon, and A. Patterson, "Motion sickness symptoms in a ship motion simulator : Effects of inside, outside, and no view," *Aviation Space Environ. Med.*, vol. 76, no. 12, pp. 1111–1118, 2005.

[70] B. Buchheit, E. Schneider, M. Alayan, and D. J. Strauss, "Motion sickness related route profiling for evaluation of the sensory conflict in real-driving studies," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2022, pp. 816–819.

[71] R. S. Johansson and J. R. Flanagan, "Coding and use of tactile signals from the fingertips in object manipulation tasks," *Nature Rev. Neurosci.*, vol. 10, no. 5, pp. 345–359, 2009.

[72] A. Greco, G. Valenza, A. Lanata, E. P. Scilingo, and L. Citi, "CvxEDA: A convex optimization approach to electrodermal activity processing," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 4, pp. 797–804, Apr. 2016.

[73] D. Bates et al., "LME4: Linear mixed-effects models using S4 classes. R package version 0.999999-0," 2012.

[74] S. Suzuki, R. Takahashi, M. Nakajima, K. Hasegawa, Y. Makino, and H. Shinoda, "Midair haptic display to human upper body," in *Proc. 57th Annu. Conf. Soc. Instrum. Control Engineers Jpn.*, 2018, pp. 848–853.

[75] C. Lehser, E. Wagner, and D. J. Strauss, "Somatosensory evoked responses elicited by haptic sensations in midair," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 10, pp. 2070–2077, Oct. 2018.

[76] I. Rakkolainen, E. Freeman, A. Sand, R. Raisamo, and S. Brewster, "A survey of mid-air ultrasound haptics and its applications," *IEEE Trans. Haptics*, vol. 14, no. 1, pp. 2–19, First Quarter 2021.

[77] D. B. Tyler, "The influence of a placebo, body position and medication on motion sickness," *Amer. J. Physiol.-Legacy Content*, vol. 146, no. 3, pp. 458–466, 1946.

[78] B. Horing et al., "Reduction of motion sickness with an enhanced placebo instruction: An experimental study with healthy participants," *Psychosomatic Med.*, vol. 75, no. 5, pp. 497–504, 2013.

[79] D. Eden and Y. Zuk, "Seasickness as a self-fulfilling prophecy: Raising self-efficacy to boost performance at sea," *J. Appl. Psychol.*, vol. 80, no. 5, Art. no. 628, 1995.

[80] N. Ljubijankić, R. Popović-Javorić, S. Šćeta, A. Šapčanin, I. Tahirović, and E. Sofić, "Daily fluctuation of cortisol in the saliva and serum of healthy persons," *Bosn. J. Basic Med. Sci.*, vol. 8, no. 2, Art. no. 110, 2008.

[81] S. Schneider, V. Brümmer, S. Göbel, H. Carnahan, A. Dubrowski, and H. K. Strüder, "Parabolic flight experience is related to increased release of stress hormones," *Eur. J. Appl. Physiol.*, vol. 100, pp. 301–308, 2007.

[82] C. Bermeitinger, "Priming," in *Psychology and Mental Health: Concepts, Methodologies, Tools, and Applications*. Hershey, PA, USA: IGI Global, 2016, pp. 42–88.

[83] O. Prasolenko, O. Lobashov, and A. Galkin, "The human factor in road traffic city," *Int. J. Automat. Control Intell. Syst.*, vol. 1, no. 3, pp. 77–84, 2015.

[84] C. Karenbach, "Ledalab-a software package for the analysis of phasic electrodermal activity," Internal report, Allgemeine Psychologie, Institut für Psychologie, Institut ftir Psychologie, Tech. Rep., 2005.

[85] S. R. Green, P. A. Kragel, M. E. Fecteau, and K. S. LaBar, "Development and validation of an unsupervised scoring system (autonomate) for skin conductance response analysis," *Int. J. Psychophysiol.*, vol. 91, no. 3, pp. 186–193, 2014.